

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **data-driven simulation and operational modelling solutions** using **Python pandas, NumPy, and scikit-learn** to optimize rail network capacity, timetabling, and passenger flow—delivering **15% improvement in schedule reliability** across enterprise transport planning dashboards.
- Led **multi-disciplinary systems engineering and safety assurance** activities for critical rail infrastructure projects, collaborating with stakeholders and delivery partners to apply **digital engineering methodologies** and ensure compliance with **Irish and UK rail standards**, regulatory frameworks, and operational safety protocols.
- Built and maintained **operational intelligence dashboards using Power BI and Tableau** to track rail asset performance, reliability metrics, and decarbonisation KPIs—enabling **real-time visibility** into network capacity, safety assurance, and passenger experience improvements across integrated rail systems.
- Supported **end-to-end rail project delivery** through detailed design assurance, UAT coordination, and post-implementation triage—managing change control for system configuration updates, stakeholder engagement, and cross-functional handovers between client, contractor, and consultant teams.
- Developed **REST API integrations and SQL-based data pipelines (PostgreSQL, MySQL, Oracle)** to connect disparate rail operational systems, enabling real-time synchronization of asset data, capacity models, and performance metrics across **interconnected infrastructure platforms**.
- Coached junior engineers and delivery teams on **best practice in digital engineering, data-driven decision-making, and professional standards**—creating training materials, Confluence documentation, and thought leadership content to support knowledge sharing and continuous improvement across multi-disciplinary rail projects.

SKILLS

Rail Operations & Digital Engineering – capacity modelling, timetabling, operational simulation, network optimization, digital engineering methodologies, safety assurance, asset performance analysis.

Data-Driven Methods & Business Intelligence – Power BI, Tableau, DAX, Power Query, data modelling, Python (pandas, NumPy, scikit-learn), statistical analysis, exploratory data analysis, A/B testing.

Systems Integration & Data Engineering – REST APIs, SQL (PostgreSQL, MySQL, Oracle), ETL pipelines, data warehousing, Python scripting, schema design, ad-hoc queries, data validation.

Cloud & DevOps Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD (Jenkins, Git), Linux, shell scripting, infrastructure automation.

Professional & Stakeholder Management – Agile/Scrum, Jira, Confluence, stakeholder engagement, UAT coordination, change management, cross-functional collaboration, proposal development, thought leadership.

Infrastructure Standards & Compliance – Irish rail standards, UK rail standards, regulatory compliance, safety protocols, audit trails, professional accreditation alignment.

Programming & Web Technologies – Java (Spring Boot), Python, Flask, FastAPI, REST APIs, React, Angular, Git version control.

EDUCATION AND TRAINING

PROJECTS

Rail Network Capacity & Operational Modelling Dashboard (HCL Technologies)

- Architected an **end-to-end operational modelling platform** using **Python (pandas, NumPy, scikit-learn)** and **Power BI** to simulate rail network capacity, timetabling scenarios, and passenger flow optimization—enabling **data-driven feasibility studies** for new and existing rail assets.
- Integrated **REST APIs** with enterprise transport planning systems to ingest real-time operational data (PostgreSQL / Oracle) and feed live metrics into **interactive Power BI dashboards** for capacity utilization, schedule reliability, and safety assurance KPIs.
- Delivered **digital engineering simulations** that improved network capacity visibility by **20%**, supporting multi-disciplinary teams (operations planners, BAs, safety assurance leads) in concept and feasibility studies across public and private sector rail clients.

Rail Asset Reliability & Decarbonisation Tracking System (HCL Technologies)

- Designed and deployed a **multi-discipline asset performance and decarbonisation dashboard** using **Tableau and Power BI**, integrating **SQL data pipelines (PostgreSQL, MySQL, Oracle)** to track rail station MEP systems, depot operations, and emissions reduction metrics aligned to **Irish and UK rail standards**.
- Led **UAT coordination and post-implementation support** with rail operations teams and delivery partners, managing change control for system configuration updates and ensuring compliance with **regulatory safety assurance protocols** and audit trail requirements.
- Enabled real-time operational reporting and thought leadership through **interactive dashboards and stakeholder engagement materials**, supporting work-winning proposals and advancing client relationships across concept design, detailed design, and assurance phases of rail infrastructure projects.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)